Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



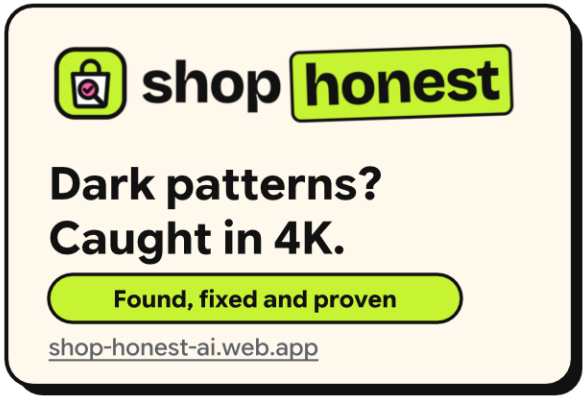
Team Details

Team name: Zero Context

Team leader name: Saubhagya Ashish

Team member: Suvojit Konar

Problem Statement: Retail & Commerce. Online stores use dark patterns (fake countdowns, items sneaked into the cart, fees revealed only at payment). India's CCPA bans 13 of them and fines platforms that use them. 26 platforms self-declared compliance, yet nobody audits the whole journey the way a shopper meets it.



Brief about the idea

THE PROBLEM

₹1,499 → ₹1,894

Our rigged test store, same earbuds: insurance sneaked into the cart, a pre-ticked membership and fees shown only at payment. The price you see isn't the price you pay.

52 of 53 top Indian apps use deceptive design
ASCI, 2024

13 dark patterns banned by India's CCPA Guidelines, 2023

26 platforms self-declared no dark patterns
Nov 2025 · one, Zepto, was fined in Aug 2026

THE IDEA

AI agents shop your store like a real customer, and fix what they find.



Every finding comes with

- Screenshot evidence
- The exact rule broken
- What it really costs
- A fix, then proof

Opportunities

1 How is it different from existing ideas?

- **Walks the whole journey** in a real browser, product to payment, instead of scanning one page or one piece of text.
- **Measures, doesn't guess:** reloads to catch restarting timers, compares cart with page, spots fees that appear only at payment.
- **Checks the MRP** against real price history in BigQuery.
- **Fixes, then proves it:** applies each fix to a copy of the store and audits it again.

2 How will it solve the problem?

- **Platforms:** a real self-audit, not a self-declaration: find, fix and prove it before the regulator does.
- **Regulators & consumer groups:** evidence at scale, with screenshots, rules and ₹ impact.
- **Shoppers:** see the real price and every trick before paying.
- **Who pays:** platforms, per monitored product; consumer groups and regulators use it free.

3 USP of the proposed solution

- **Proof, not opinions:** every claim cites a measured page element and a second AI double-checks it; unsure claims go to a human.
- **Speaks money and law:** "+₹395 per order", the rule broken in 7 regions, and a fix for each finding.
- **Honest about limits:** stops at sign-in walls and bot protection; a blocked page is "not checked", never "clean".

Proof it works: **9/9** tricks caught · **0** false alarms · **9/9** fixes verified · **0.959 F1** zero-shot · **1** real store caught

List of features offered by the solution

01

Live agent walkthrough

Watch agents shop in a real browser, narrating every step as they go.

02

Measured checks

Restarting timers, frozen stock, sneaked items, pre-ticked boxes, hidden fees.

03

Price Historian

BigQuery price history exposes "MRPs" the product never sold at.

04

ADK agent graph

Gemini reads each screen; an Adjudicator agent checks the evidence with tools.

05

Fix it + verify

A fix for each finding; watch the score move, then verify it on a copy.

06

The real bill

Price seen vs. price charged, the true saving, overcharge at scale.

07

Law + CCPA self-audit

The rule broken in India and 6 more regions; a 13-pattern CCPA self-audit PDF.

08

Reports for any store

Any public store, logged out; a blocked page says "not checked", never "clean".

Process flow diagram or Use-case diagram

THE JOURNEY a logged-out shopper, in a real browser

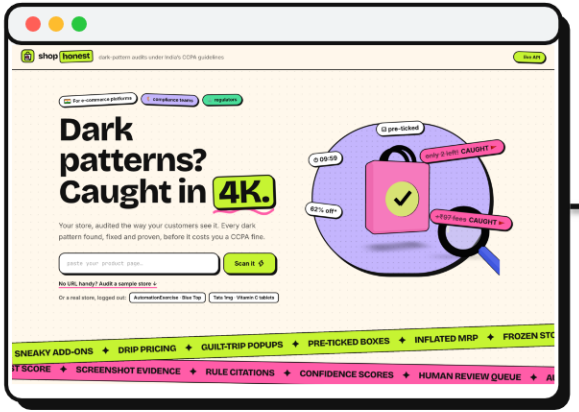


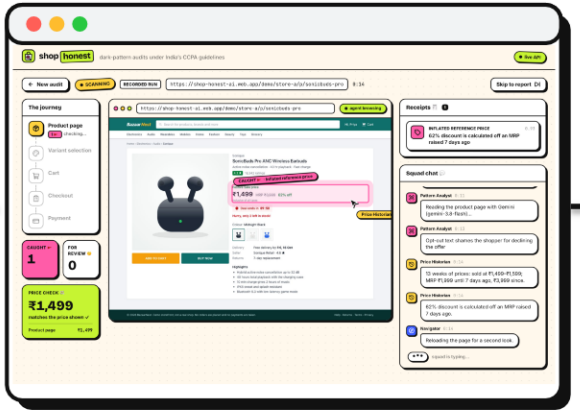
ON EVERY SCREEN an ADK agent graph: measure first, then Gemini judges, then an agent re-checks with tools



Report trust score · the real bill · law in 7 regions · CCPA self-audit · PDF → Fix it: projected, then verified on a copy

Wireframes/Mock diagrams of the proposed solution (optional)







1 Paste a URL or pick a store

Sample stores, or any public store, logged out.

2 Watch the agents shop, live

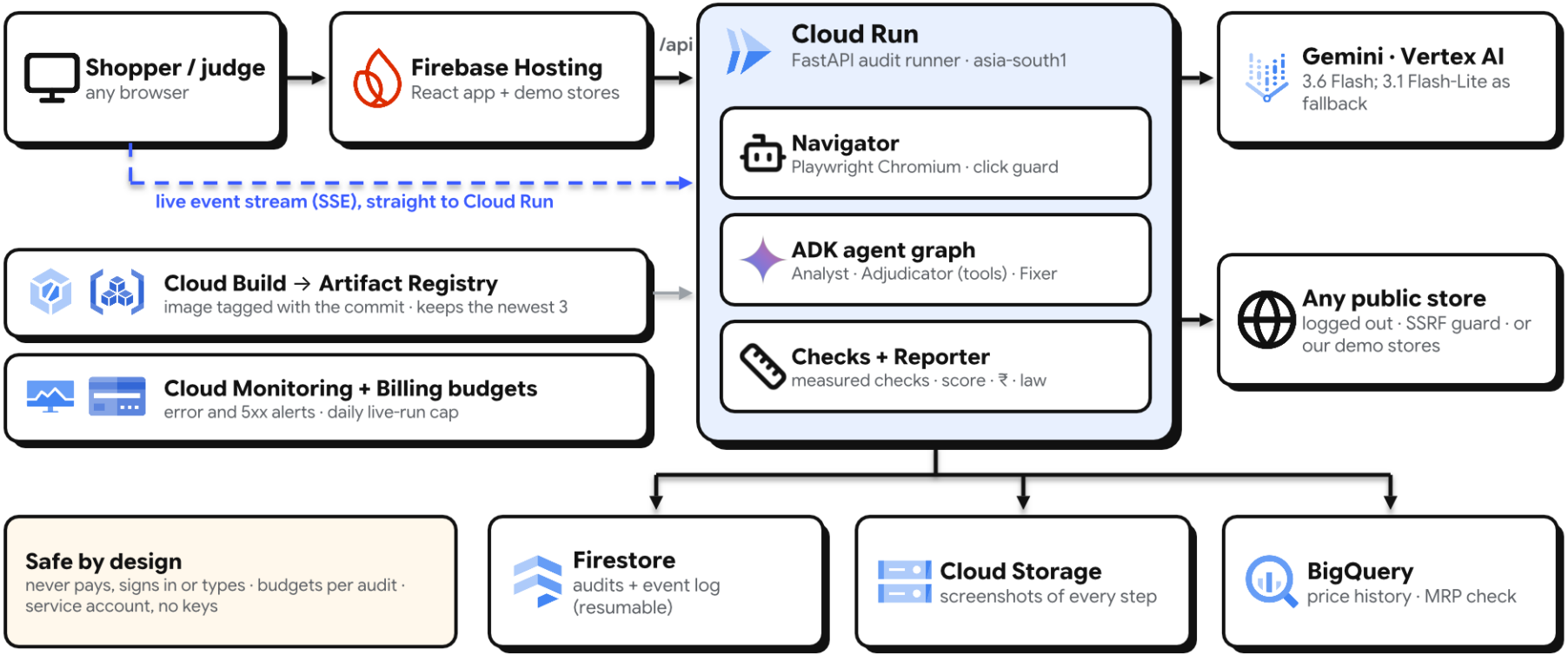
Live browser, journey, receipts so far and the agents' chat.

3 Get the report, then Fix it

Trust score, the real bill, fine risk and evidence. Then Fix it and verify.

Not a mock: this is the live product. Try it at shop-honest-ai.web.app · every run is a live audit with Gemini

Architecture diagram of the proposed solution



Technologies to be used in the solution

AI	**Gemini 3.6 Flash** on Vertex AI	**Google ADK** agent graph + tools	**Gemma 3 4B** local dev, via Ollama
Google Cloud	**Cloud Run** API + browsers	**Firestore** audits, events	**Cloud Storage** screenshots
	BigQuery price history	**Cloud Build** container builds	**Artifact Registry** images
	Cloud Monitoring alerts, uptime	**Cloud Billing** budgets	**Firebase Hosting** web app + CDN
App	**Python 3.12** + uv	**FastAPI** API + live stream	**Playwright** headless Chromium
	Pydantic typed model output	**React 19** dashboard	**TypeScript** end to end
	Vite build	**Tailwind CSS** styling	**three.js** 3D hero

Estimated implementation cost (optional)

PER LIVE AUDIT

≈ ₹2

a full journey: 5 steps, about 30 seconds

Gemini 3.6 Flash ~21k in + 5k out tokens	**₹1.5–2**
Cloud Run ~30 s on 2 vCPU	**< ₹0.5**
Firestore · Storage · BigQuery events, screenshots, prices	**paise**
Hosting · builds · alerts Firebase Hosting, Cloud Build	**≈ ₹0**

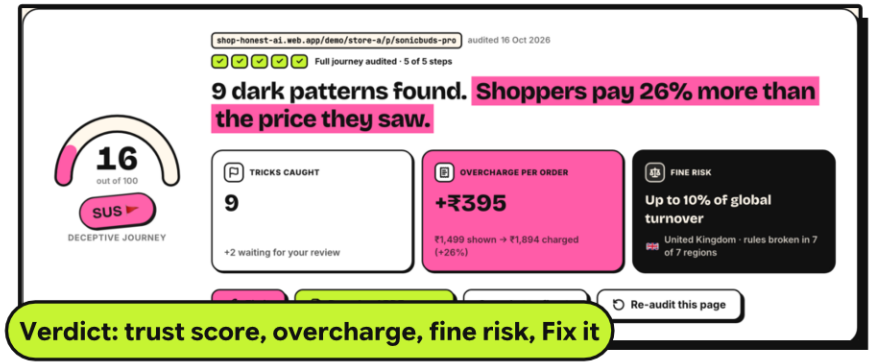
MONTHLY ESTIMATE live audits only; idle costs ≈ ₹0 (Cloud Run scales to zero)

1,000 audits / month	10,000 audits / month	1 lakh audits / month
₹2,000	**₹20,000**	**₹2 lakh**
a pilot with one platform	daily checks on 300+ products	a regulator's sweep

GUARDRAILS so a bill can't surprise anyone

- 100 live runs a day, site-wide
- ≤ 400k tokens, 40 calls per audit
- 6-minute limit per audit
- Budget alerts at ₹2,000 / month
- Keeps only the 3 newest images
- Scales to zero when idle

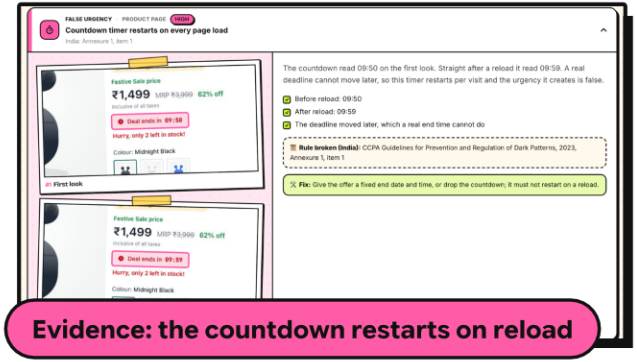
Snapshots of the prototype



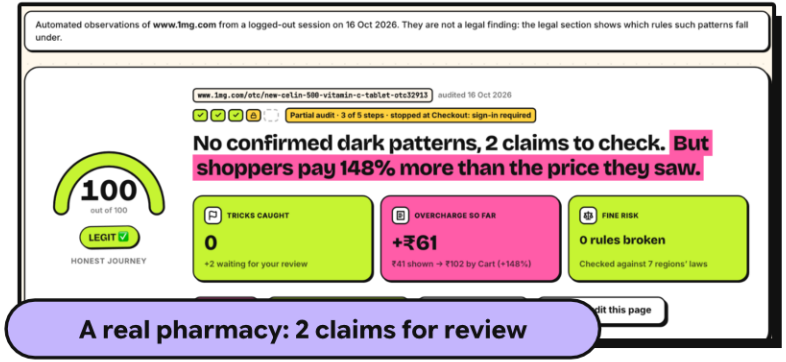
Verdict: trust score, overcharge, fine risk, Fix it



The bill: ₹1,499 shown, ₹1,894 charged

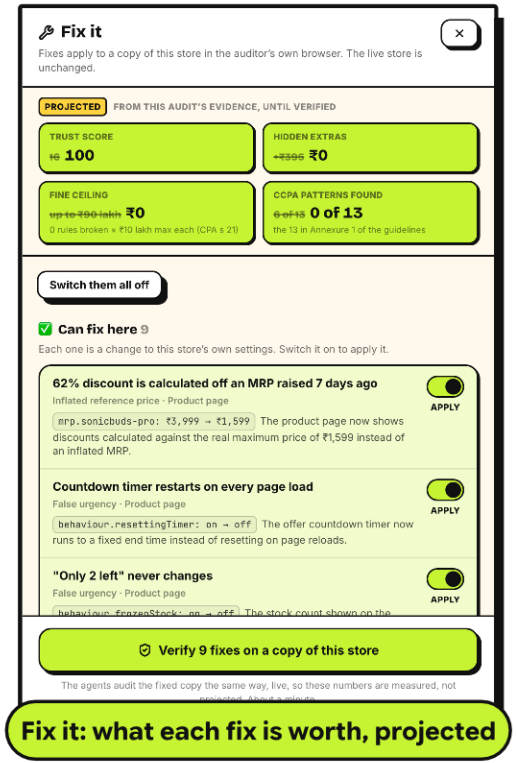


Evidence: the countdown restarts on reload

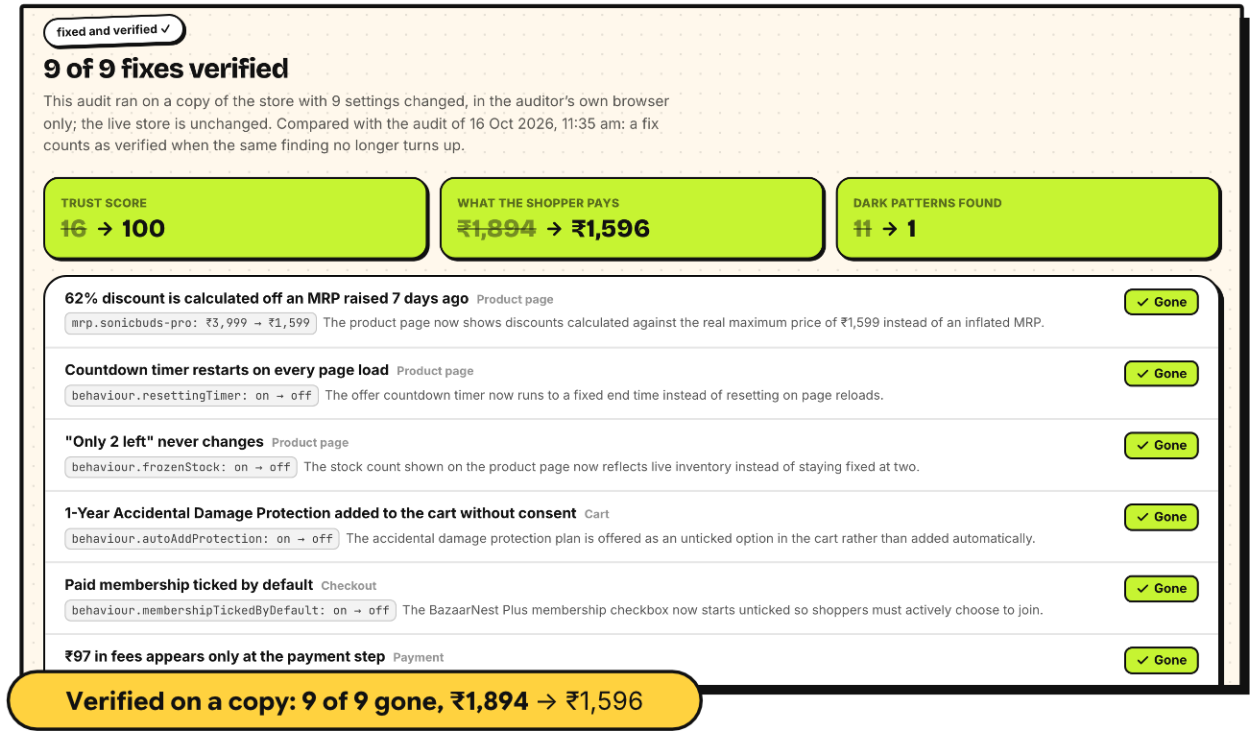


A real pharmacy: 2 claims for review

Snapshots of the prototype: Fix it



Fix it: what each fix is worth, projected



Verified on a copy: 9 of 9 gone, ₹1,894 → ₹1,596

Prototype Performance report/Benchmarking

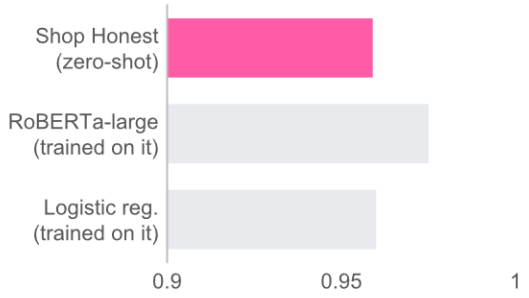
PLANTED TRICKS live runs, 2 per store

9/9 **tricks caught · Store A**
the borderline claim went to human review

0 **false alarms · Store B**
the honest control store

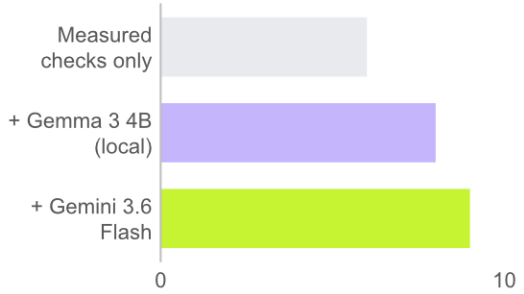
6/6 **caught before the wall · Store C**
then stops honestly at sign-in

PUBLIC DATASET 2,356 texts, F1



0.997 **precision**
3 false alarms in 2,356 texts. Our model never saw this dataset; the others were trained on it.

EVERY LAYER COUNTS Store A recall



~27 s **per audit**
2.5 Pro, 3.1 Pro and 3.8 Flash also scored 9/9 but were slower, so 3.6 Flash runs both tiers.

Also tested: 224 backend + 56 web tests · Fix it: 9/9 fixes verified on a copy · every event checked against the API contract · pre-flight 6/6

Tested on the real platforms: 6 of the 26 self-declared



CART

Tata 1mg

Reached the cart, then its sign-in popup. A ₹12 handling charge first shown in the cart: sent for review.



CART

Flipkart

Reached the cart: a ₹10 platform fee the product page didn't show, sent for review. Later visits got an error page.



SOLD OUT

Blinkit

The product was out of stock, so the agent stopped instead of adding a similar one.



EMPTY

Zepto

Fined by the CCPA in Aug 2026. Served our browser an empty page: reported as not checked, never as clean.



REFUSED

Myntra

Refused the connection from an automated browser. Reported as such; the agent never gets around it.



BOT WALL

Meesho

Bot protection stopped the agent on the first page. Reported as such, never as clean.

Never once wrongly called clean. Outside audits get blocked, so the audit belongs on the platform's side, before release.

Tested on 21 real websites, 82 audits: one store caught



CONFIRMED

1 store caught

A D2C fashion brand: a ₹199 membership put in the cart unasked (review) and a decline worded to shame (confirmed).



CART

8 reached the cart

Seven went on until checkout asked them to sign in; one had no way to checkout.



REVIEW

3 sent for review

A ₹12 fee first shown in the cart; unsourced "N people bought this" claims. Flagged, never asserted.



NO CART

6 stopped at the product

Out of stock, an error page, a build-your-own product, or a site that sells only by enquiry.



BLOCKED

3 blocked us

Bot protection, empty or error pages, a refused connection. Reported as not checked, never as clean.



15–17 OCT

Logged out, never paying

Marketplaces, a pharmacy, quick commerce and D2C brands, in India and the US. A fresh browser every time.

Stores aren't named: automated observations of a logged-out session. Raw reports on request.

Responsible by design

The agent will never

- Pay, buy now or place an order
- Sign in, register or subscribe
- Type into any field
- Accept cookies or terms (it declines or closes)
- Get around a CAPTCHA or bot protection

Findings stay honest

- **Grounded:** every claim must cite an element measured on the page.
- **Double-checked:** a second agent re-checks the evidence with tools.
- **Unsure? Human review,** never asserted as fact.
- **Real stores:** layout-based findings go to review; reports say "not a legal finding".
- **Partial audits** say exactly which steps weren't seen.

Guardrails: SSRF guard · ad networks blocked · per-audit budgets · daily cap · Fix it runs only on a copy, in our own browser

Additional Details/Future Development (if any)

NEXT

The 26, audited

All 26 self-declared platforms, logged out, each finding checked by a person.

NEXT

Logged-in journeys

Test accounts in Secret Manager, so checkout and payment on real stores get audited too.

NEXT

Scheduled monitoring

Nightly re-audits with Cloud Scheduler, and a CI check that fails a deploy adding a trick.

LATER

Shopper extension

A browser extension that flags tricks while you shop, with the real price.

LATER

Regulator dashboard

BigQuery + Looker Studio trends across platforms, ready for CCPA filings.

LATER

Apps & Indian languages

Mobile app journeys, and pages in Hindi and other Indian languages.

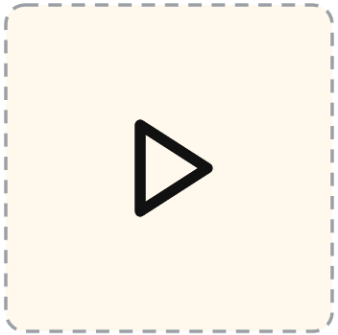
Provide links to your:

GitHub Public Repository

github.com/saubhagyaashish/shop-honest

Code, docs, design decisions and evals

Demo Video Link (3 Minutes)



youtu.be/_mG4SobZ9E0

A 2:54 product walkthrough

Final Product Link

shop-honest-ai.web.app

Sample stores, live runs and real stores

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

Found, fixed and proven, before it costs you a CCPA fine.

shop-honest-ai.web.app

Team Zero Context · Saubhagya Ashish · Suvojit Konar

